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- 1 A factory produces small bottles of natural spring water. Two different machines, X and Y , are used to fill empty bottles with the water. A quality control engineer checks the volumes of water in the bottles filled by each of the machines. He chooses a random sample of 60 bottles filled by machine X and a random sample of 75 bottles filled by machine Y . The volumes of water, x and y respectively, in millilitres, are summarised as follows.

$$\Sigma x = 6345 \quad \Sigma (x - \bar{x})^2 = 243.8 \quad \Sigma y = 7614 \quad \Sigma (y - \bar{y})^2 = 384.9$$

$\bar{x}$ and $\bar{y}$ are the sample means of the volume of water in the bottles filled by machines X and Y respectively.

Find a 95% confidence interval for the difference between the mean volume of water in bottles filled by machine X and the mean volume of water in bottles filled by machine Y . [6]

$$(\bar{x} - \bar{y}) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}} \quad \left\{ \begin{array}{l} (x - \bar{x})^2 \\ \rightarrow \Sigma x^2 - \frac{(\Sigma x)^2}{n} \end{array} \right.$$

$$\bar{x} = \frac{6345}{60} = 105.75 \quad \bar{y} = \frac{7614}{75} = 101.52$$

$$s_x^2 = \frac{1}{59} (243.8) = 4.1322$$

$$s_y^2 = \frac{1}{74} (384.9) = 5.201$$

$$z_{\frac{\alpha}{2}} = z_{0.025} \quad \text{where } 100(\alpha - 1) = 95$$

$$C.I \rightarrow (105.75 - 101.52) \pm 1.96 \sqrt{\frac{4.1322}{60} + \frac{5.201}{75}}$$

$$\rightarrow 4.23 \pm (1.96 \times 0.3718)$$

$$\rightarrow [3.50, 4.96]$$

- 2 The number of breakdowns on a particular section of road is recorded each day over a period of 90 days. It is suggested that the number of breakdowns follows a Poisson distribution with mean 3.5. The data is summarised in the table, together with some of the expected frequencies resulting from the suggested Poisson distribution.

Number of breakdowns per day	0	1	2	3	4	5	6	7	8 or more
Observed frequency	0	5	13	17	21	16	9	5	4
Expected frequency	2.718	9.512	16.646	19.421	16.993	11.895	6.939	3.469	2.407

- (a) Complete the table.

[2]

$$E(3) = e^{-3.5} \times \frac{3.5^3}{3!} \times 90 = 19.421$$

$$E(6) = e^{-3.5} \times \frac{3.5^6}{6!} \times 90 = 6.939$$

- (b) Carry out a goodness of fit test, at the 10% significance level, to determine whether or not $Po(3.5)$ is a good fit to the data.

[6]

$H_0: Po(3.5)$ is a good fit to the data

$H_1: Po(3.5)$ is not a good fit for the data

Combine first two and last two columns
so that Expected values ≥ 5

	0-1	2	3	4	5	6	7 or more
obs	5	13	17	21	16	9	9
exp	12.23	16.646	19.421	16.993	11.895	6.939	5.876

$$\text{Test statistic} \rightarrow \sum \frac{(O - E)^2}{E}$$

$$\rightarrow \frac{(5 - 12.23)^2}{12.23} + \dots$$

$$\rightarrow 4.274 + 0.798 + 0.3018 + 0.949 + 1.417 + 0.612 + 1.661 = 10.0088$$

$$\text{degrees of freedom} = n - \underbrace{\nu}_{\substack{\text{no estimated} \\ \text{parameters}}} - 1$$

$$7 - 0 - 1 = 6$$

$$\chi^2_6(0.90)$$

critical value $\rightarrow 10.64$

$$10.0088 < 10.64 \rightarrow \text{accept } H_0$$

insufficient evidence to suggest that $P_0(3.5)$ is not a good fit for the data

- 3 Toby has a bag which contains 6 red marbles and 3 green marbles. He randomly chooses 3 marbles from the bag, without replacement. The random variable X is the number of red marbles that Toby obtains.

(a) Find the probability generating function of X .

[3]

$$P(X=0) = \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} = \frac{1}{84}$$

$$P(X=1) = \frac{6}{9} \times \frac{3}{8} \times \frac{2}{7} \times \frac{3!}{2!} = \frac{3}{14}$$

$$P(X=2) = \frac{6}{9} \times \frac{5}{8} \times \frac{3}{7} \times \frac{3!}{2!} = \frac{15}{28}$$

$$P(X=3) = \frac{6}{9} \times \frac{5}{8} \times \frac{4}{7} = \frac{5}{21}$$

$$G_X(t) = \frac{1}{84} + \frac{3}{14}t + \frac{15}{28}t^2 + \frac{5}{21}t^3$$

Ling also has a bag which contains 6 red marbles and 3 green marbles. He randomly chooses 2 marbles from his bag, without replacement. The random variable Y is the number of red marbles that Ling obtains. It is given that the probability generating function of Y is $\frac{1}{12}(1+6t+5t^2)$.

The random variable Z is the total number of red marbles that Toby and Ling obtain.

(b) Find the probability generating function of Z , expressing your answer as a polynomial in t .

[3]

$$G_X(t) = \frac{1}{84} (1 + 18t + 45t^2 + 20t^3)$$

$$G_Z(t) = \frac{1}{84} \times \frac{1}{12} (1 + 6t + 5t^2)(1 + 18t + 45t^2 + 20t^3)$$

$$\rightarrow \frac{1}{1008} (1 + 18t + 45t^2 + 20t^3 + 6t + 108t^2 + 270t^3 + 120t^4 + 5t^2 + 90t^3 + 225t^4 + 100t^5)$$

$$= \frac{1}{1008} (1 + 24t + 158t^2 + 380t^3 + 345t^4 + 100t^5)$$

$$\frac{1}{1008} (1 + 158t^2 + 24t + 380t^3 + 345t^4 + 100t^5)$$

- (c) Use the probability generating function of Z to find $\text{Var}(Z)$.

$$\text{Var}(X) = G''_X(1) + G'_X(1) - (G'_X(1))^2 \quad [4]$$

$$G'_X(t) = \frac{1}{1008} (24 + 316t + 1140t^2 + 1380t^3 + 500t^4)$$

$$G'_X(1) = \frac{3360}{1008} = \frac{10}{3}$$

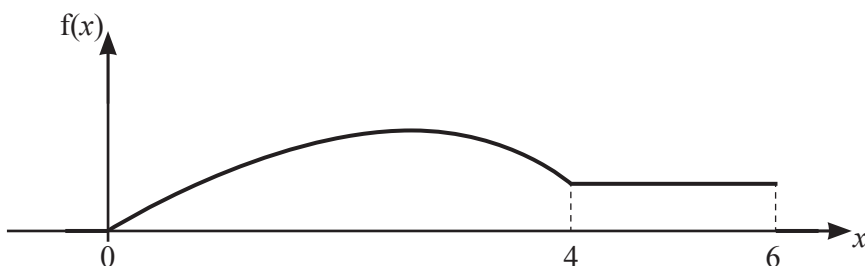
$$G''_X(t) = \frac{1}{1008} (316 + 2280t + 4140t^2 + 2000t^3)$$

$$G''_X(1) = \frac{26}{3}$$

$$\therefore \text{Var}(X) = \frac{26}{3} + \frac{10}{3} - \left(\frac{10}{3}\right)^2$$

$$= \frac{8}{9}$$

4



The diagram shows the continuous random variable X with probability density function f given by

$$f(x) = \begin{cases} \frac{1}{128}(4ax - bx^3) & 0 \leq x \leq 4, \\ c & 4 \leq x \leq 6, \\ 0 & \text{otherwise,} \end{cases}$$

where a , b and c are constants.

The upper quartile of X is equal to 4.

(a) Show that $c = \frac{1}{8}$ and find the values of a and b .

[4]

$$c \times (6-4) = 0.25$$

$$c = \frac{1}{8} \quad \text{as total amount of area of PDF after UQ} \rightarrow 0.25$$

$$\text{at } x=4 \quad f(x) = \frac{1}{8}$$

$$\frac{1}{128}(16a - 64b) = \frac{1}{8} \rightarrow \boxed{a - 4b = 1}$$

$$\frac{1}{128} \int_0^4 (4ax - bx^3) dx = 0.75$$

$$\left[2ax^2 - \frac{bx^4}{4} \right]_0^4 = 96$$

$$32a - 64b = 96 \rightarrow a - 2b = 3$$

$$- \quad \underline{a - 4b = 1}$$

$$2b = 2$$

$$\boxed{a = 5}$$

$$\boxed{b = 1}$$

(b) Find the exact value of the median of X .

[3]

median in range $\rightarrow 0 \leq x \leq 4$

$$\frac{1}{128} \int_0^m 20x - x^3 dx = 0.5$$

$$\frac{1}{128} \left[10x^2 - \frac{x^4}{4} \right]_0^m = 0.5$$

$$10m^2 - \frac{m^4}{4} = 64$$

$$m^4 - 40m^2 + 256 = 0$$

$$(m^2 - 32)(m^2 - 8) = 0 \quad m = \sqrt{8}$$

$$m = 2\sqrt{2}$$

(c) Find $E(\sqrt{X})$, giving your answer correct to 2 decimal places.

[3]

$$E(\sqrt{x}) \rightarrow \int \sqrt{x} f(x) dx$$

$$\frac{1}{128} \int_0^4 20x^{3/2} - x^{5/2} dx + \frac{1}{8} \int_4^6 x^{1/2} dx$$

$$\frac{1}{128} \left[8x^{5/2} - \frac{2}{9}x^{7/2} \right]_0^4 + \frac{1}{8} \left[\frac{2}{3}x^{3/2} \right]_4^6$$

$$\rightarrow \frac{1}{128} \left[\frac{1280}{9} \right] + \frac{1}{8} [4.4646] = \boxed{1.67}$$

- 5 A company is deciding which of two machines, X and Y , can make a certain type of electrical component more quickly. The times taken, in minutes, to make one component of this type are recorded for a random sample of 8 components made by machine X and a random sample of 9 components made by machine Y . These times are as follows.

Machine X	4.0	4.6	4.7	4.8	5.0	5.2	5.6	5.8	
Machine Y	4.5	4.9	5.1	5.3	5.4	5.7	5.9	6.3	6.4

The manager claims that on average the time taken by machine X to make one component is less than that taken by machine Y .

- (a) Carry out a Wilcoxon rank-sum test at the 5% significance level to test whether the manager's claim is supported by the data. [6]

X		Rank	Y		Rank	
4.0	1		4.5	2		$m \leq n$
4.6	3		4.9	6		
4.7	4		5.1	8		$H_0: m_x = m_y$
4.8	5		5.3	10		
5.0	7		5.4	11		$H_1: m_x < m_y$
5.2	9		5.7	13		
5.6	12		5.9	15		
5.8	14		6.3	16		
			6.4	17		
		55			98	

$$\begin{aligned}
 W &\sim \min(k_m, n(m+n+1) - k_m) \\
 &\sim \min(55, 89) \\
 &= 55
 \end{aligned}$$

critical value $\rightarrow m=8, n=9, 5\%$. one-tailed
 $\rightarrow 54$

$55 > 54 \rightarrow$ accept H_0
 $\rightarrow$ insufficient evidence to support manager's claim

- (b) Assuming that the times taken to produce the components by the two machines are normally distributed with equal variances, carry out a t -test at the 5% significance level to test whether the manager's claim is supported by the data. [9]

$$H_0: \mu_x = \mu_y \quad H_1: \mu_x < \mu_y$$

$$\sum x = 39.7$$

$$\sum y = 49.5$$

$$\sum x^2 = 199.33$$

$$\sum y^2 = 275.47$$

$$S_p^2 = \frac{S_{xx} + S_{yy}}{n_x + n_y - 2}$$

$$S_{xx} = \frac{199.33 - \frac{(39.7)^2}{8}}{8} = 2.31875$$

$$S_{yy} = \frac{275.47 - \frac{(49.5)^2}{9}}{9} = 3.22$$

$$S_p^2 = \frac{3.22 + 2.31875}{15} = 0.36925$$

$$T = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{\sqrt{S_p^2 \left(\frac{1}{n_x} + \frac{1}{n_y} \right)}}$$

$$= \frac{\frac{39.7}{8} - \frac{49.5}{9}}{\sqrt{0.36925 \left(\frac{1}{8} + \frac{1}{9} \right)}} = -1.820$$

$$\text{critical value} \rightarrow t_{0.05, 15} = -1.753$$

$$-1.820 < -1.753 \rightarrow \text{reject } H_0$$

Question 5(c) is printed on the next page.

sufficient evidence that ^{mean} time taken
by machine X is less than
mean time for machine Y

- (c) In general, would you expect the conclusions from the tests in parts (a) and (b) to be the same? Give a reason for your answer. [1]

t-test $\rightarrow$ assumes a normal distribution
 $\rightarrow$ equal variances
 $\rightarrow$ but this may not be true
 $\rightarrow$ therefore the two results
may not be the same

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